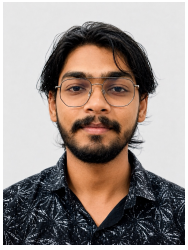
Ayush Kumar Gupta

Curriculum Vitae

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ayushhgupta1



Research Interests

My research interests lie in computer vision and applied AI, particularly in adapting foundation models for real-world problems. My work spans unsupervised segmentation and biomass estimation from satellite imagery, as well as language-guided object grounding for robotic manipulation. I am interested in visual representation learning, segmentation, and grounding, and in developing methods that make pretrained models more effective for specific tasks and I am looking to pursue a PhD building in this direction..

Education

2023 – Present **Indian Institute of Technology Dharwad**, B.Tech, Electrical Engineering
CPI: 8.46/10

2023 **Class XII (CBSE)**, 92.4%

2021 **Class X (CBSE)**, 94.0%

Scholastic Achievements

NTSE Scholar (National Talent Search Examination, NCERT, National Level)
IOQM'21: Secured State Rank under 300 in Indian Olympiad Qualifier in Mathematics

Research Experience

Aug 2026 – Ongoing **Vision Lab, IIT Dharwad** | Mentor: Dr. Shashaank A. Mattur, Assistant Professor, EE
Ambiguity-Aware Language-Guided Object Grounding for Robotic Pick-and-Place

- Developed an object-centric referring segmentation framework — a 16M-parameter Transformer-based grounder over frozen SAM and CLIP features, geometric attributes, and pairwise object relationships — that identifies target objects from natural-language instructions and predicts whether to act, reject, or seek clarification when multiple objects match.
- Constructed D4 (303K phrase–image pairs across 19,994 COCO images) and extended training with LVIS queries to improve generalization across object categories.
- Built an ambiguity-resolution mechanism (duplicate-mask grouping, cardinality estimation, scene-grounded clarification questions) that raised simulated-user success on ambiguous referring expressions from 3.0% to 80.6% at roughly one question per request, and reaches 0.989 AUROC for ambiguity detection with 83.6% single-target accuracy on LVIS.
- **Ongoing:** building an end-to-end interactive assistant with iterative clarification and language-guided placement-point estimation.

Aug 2025 – Jan 2026 **IIT Dharwad** | Mentor: Dr. Shashaank A. Mattur, Assistant Professor, EE
Semi-Automated Estimation of Above-Ground Biomass in Agroforestry

- **Presented at IEEE IGARSS 2026, Washington D.C.; currently awaiting publication in the conference proceedings.**
- Built a pipeline leveraging the Segment Anything Model (SAM) for tree crown and shadow segmentation from very high-resolution (VHR) satellite imagery, enabling scalable above-ground biomass estimation without dense field measurements.
- Designed a post-processing framework using shadow geometry to estimate tree height and biomass relative to a reference tree, reducing dependence on manual field surveys.
- Enabled faster, large-scale agroforestry biomass assessment by automating key stages of the estimation workflow, reducing the time and manual effort required compared to conventional field-based surveys.

Jan 2026 – May 2026

IIT Dharwad | Mentor: Dr. Shashaank A. Mattur, Assistant Professor, EE
Unsupervised Segmentation of High-Resolution Agroforestry Plots
Extended the above IGARSS biomass-estimation work to remove its dependence on labeled data, by developing a fully unsupervised satellite image segmentation pipeline.

- Used DINOv2 transformer embeddings to generate coarse segmentation maps requiring no labeled training data, refined with Dense CRF post-processing for spatial consistency across tree canopy boundaries.
- Performed instance separation via distance transform combined with a flooding-based algorithm for individual tree detection, addressing high intra-class variability and low inter-class separability in dense canopy cover.

Work Experience

Jun 2025 – Jul 2025

Research Intern, FIST-TBI (Technology Business Incubator), IIT Patna
Guide: Dr. Maheshkumar H. Kolekar, Associate Professor, EE, IIT Patna
Real-Time Pothole (and Obstacle) Detection and Pre-Warning System – Certificate

- Worked on developing a real-life road safety solution involving image-based pothole detection and alert generation.
- Led the end-to-end development of a smart pothole detection and alert system aimed at real-time road hazard identification for safer driving experiences.
- Designed and developed a real-time pothole detection and pre-warning system using YOLOv12 for object detection and Deep SORT for object tracking and unique pothole counting.
- Built a complete prototype using Raspberry Pi, ultrasonic sensors, and GPS modules for on-road deployment and real-time alerts.
- Optimized the detection pipeline to reduce frame lag and eliminate overcounting using tracking-based logic and frame-skipping techniques.
- Transitioned from an initial Arduino-based model to a fully integrated, edge-deployable Raspberry Pi system for improved performance.
- Explored advanced sensor integration with LiDAR and IMU to enhance depth sensing and road profile detection.

Jun 2026 – Jul 2026

AI Analyst, Perceptive Analytics | Remote
AI-Driven Financial Summarization – Certificate

- Contributed to the development of an AI-driven financial summarization application utilizing Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) architectures.
- Primarily focused on refactoring and optimizing the existing backend codebase to enhance system performance and scalability, including leveraging established Redis caching layers.
- Additionally involved in managing application deployment via DigitalOcean and finalizing comprehensive technical documentation for the system.

Key Projects

Jan 2026 – Apr 2026

Real-Time Multimodal Emotion Sensing System (Edge Deployment)
Electronic Design Lab — Group Project

- Developed a multimodal emotion recognition system integrating text, speech, and facial inputs.
- Used MiniLM for text-based emotion classification, MobileNetV3 for real-time facial emotion detection, and Vosk for speech-to-text processing.
- Designed a multimodal fusion pipeline to compute a real-time "Happiness Index" combining all three modalities.
- Optimized models for Raspberry Pi 4 edge deployment, handling training, evaluation, optimization, and system-level integration.

Course Project

Deep Reinforcement Learning Framework for Multi-Exchange Spatial Arbitrage
Neural Networks & Deep Learning

- Designed a simulation testbed for low-latency cross-exchange arbitrage, modeling the market as a 9-node asset-exchange graph and using Graph Convolutional Networks (GCNs) for spatial feature extraction.
- Implemented a Dueling DQN over a 19-dimensional action space and built a multi-threaded WebSocket pipeline for real-time simulation, telemetry, latency reduction, routing optimization, and distributed inventory management.

Technical Skills

Languages	C, C++, Python
ML/Data	PyTorch, Deep Learning, GNN, CNN, Random Forests, Pandas, NumPy, scikit-learn, ETL, Reinforcement Learning
CS Core	Data Structures and Algorithms, OOP, Design Patterns, DBMS

Relevant Coursework

Machine Learning & Computer Vision

- Neural Networks and Deep Learning (EE608T, **AA**)
- Neural Networks and Deep Learning Lab (EE601L, **AA**)
- Computer Vision and Digital Image Processing (EE630T, **AB**)

Mathematics

- Complex Analysis (MA201, **AA**)
- Linear Algebra (MA102, **BB**)
- Introduction to Probability (EE221, **BB**)

Computer Science & Systems

- Data Structures and Algorithms (CS201, **BB**)
- Microprocessors and Microcontrollers (EE301T, **AB**)
- Microprocessors and Microcontrollers Lab (EE302L, **AA**)

Research

- Research and Development Project I & II (EE301P, EE302P, **AA**)

Grade scale: ***AA*** *= 10/10,* ***AB*** *= 9/10,* ***BB*** *= 8/10*

Positions of Responsibility

Mentor, ISMP Program
Mentored junior B.Tech. students in academic matters, campus life, and extracurricular activities, helping them navigate university life and make the most of available opportunities.

Volunteer, PARSEC 5.0 (Annual Technical Fest, IIT Dharwad)
Contributed to the organisation and smooth execution of technical events, assisting with event coordination and on-ground operations.

Hobbies & Interests

- **Poetry Writing:** Urdu/Hindi poetry and Mushaira.
- Represented IIT Dharwad at Inter IIT Cult Meet 7.0, IIT Patna, in Poetry and Mushaira, and Inter IIT Cult Meet 8.0, IIT Kanpur, in Mushaira.
- Secured **3rd position** in Poetry Writing (Alfaz).